

From “Balance” to Dynamics

A Control-Theoretic Formalization of the Metacognitive
Balance Perspective of Han & Gu (2025)

Working note, prepared for the collaboration with Prof. Mihwa Han

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Abstract

Han & Gu (2025) argue that cognitive and social functioning in schizophrenia is governed less by *the order* in which self-awareness and context/other-awareness are trained than by their *concurrent co-regulation*, and they picture this balance as movement within an “adaptive corridor” policed by immediate “pivots.” This note takes their verbal framework at face value and asks a narrow question: *if we write these claims down as equations, does the resulting system actually reproduce the clinical phenomena they describe?* I formalize interpretation as a probabilistic belief, metacognition as a control law acting on that belief, and the self–context referencing system as a coupled dial. A minimal single-scenario realization then reproduces, without extra assumptions, four described phenomena: consolidation of a threat belief under over-internalization, erratic cue-tracking under over-externalization, jumping-to-conclusions as premature confidence, and the rescue of a consolidating belief by a within-session pivot. Two further studies show that the chapter’s own “minimal indices” (confidence–outcome congruence and tolerance of ambiguity) separate the regimes, and that concurrent co-regulation quantitatively outperforms a gated “self-then-context” sequence. Every modeling decision is anchored to the specific sentence in Han and Gu (2025) that motivated it, and every mathematical object is glossed in plain language. This is offered as an *interpretation* of the chapter, not a competing theory.

1 Purpose, and how to read this note

The balance-oriented perspective of Han and Gu (2025) is stated as a *conceptual* direction; the authors explicitly reserve formalization for later work. This note is a first attempt at that formalization, written so that it can be read from two sides at once. For the modeler, the goal is a system of differential equations that is simple enough to simulate and inspect. For the psychologist—and especially for Prof. Han, whose chapter this formalizes—the goal is that each equation be traceable to a sentence she wrote, and that no symbol go unexplained. Wherever an equation appears, a shaded blue box restates it in ordinary language.

Think of this note as a translation. The chapter is written in the language of clinical observation; I am re-writing the *same claims* in the language of mathematics, the way one might re-score a piece of music for a different instrument. The tune should be recognizably yours. If any box does not sound like what you meant, that is exactly the kind of mismatch worth catching early.

Notation at a glance. A handful of symbols recur. Table 1 lists them in plain terms.

2 From the text to the equations

I build the model in the order the chapter builds its argument: first what a metacognitive judgment *is*, then that it is *controlled*, then that the control has a *self-context dial*, and finally

Table 1: The symbols used in this note, in ordinary language.

Symbol	Reads as	Plain meaning
$L(t)$	“log-odds of threat”	a single number for the belief: 0 = truly unsure (50/50), positive = leaning “it’s about me / threatening,” negative = “benign.”
$\sigma(L)$	“sigmoid of L ”	an S-shaped squashing function turning L into a probability in $[0, 1]$.
$b(t) = \sigma(L)$	“belief”	the probability the person currently assigns to the threat reading.
$\alpha(t)$	“self-weight”	a dial in $[0, 1]$: how much inner signals count vs. outer cues.
$\frac{dL}{dt}$	“rate of change of L ”	how fast the belief is moving <i>right now</i> .
κ	“leak”	a gentle pull back toward “not sure yet” (guards over-commitment).
γ, α_0	“gain, set-point”	how strongly, and toward what habitual lean, the dial returns.
$u(t)$	“control input”	the therapist’s nudge (a pivot / micro-cycle).

what happens when the dial is stuck or is nudged.

2.1 Interpretation is a probability, not a verdict

FROM HAN & GU (2025), P. 1 ... *the capacity to consistently integrate and interpret thoughts, feelings, and bodily sensations in light of others’ intentions, prevailing social norms, and situational cues, and to regulate responses accordingly.*

Two things are asserted here. First, a metacognitive output is an *interpretation* formed by *integrating* internal material (thoughts, feelings, sensations) with external material (others’ intentions, norms, cues). Second, that interpretation feeds *regulation*. I therefore split the construct into an output (the interpretation) and a controller (the regulation), and I make the output a *probability* rather than a fixed answer, because the chapter repeatedly treats premature certainty as pathological:

FROM HAN & GU (2025), P. 5 *Decisions are often driven by automatic feelings (“feeling = fact”) without explicit reflection, leading to premature closure (JTC) before controlled re-appraisal comes online.*

For a single ambiguous social event with two readings— h_1 (benign) and h_2 (threat / self-referential)—I represent the interpretation as the posterior probability of the threat reading, written on the log-odds scale:

$$L(t) = \log \frac{P(h_2 \mid \text{evidence up to } t)}{P(h_1 \mid \text{evidence up to } t)}, \quad b(t) = \sigma(L(t)) = \frac{1}{1 + e^{-L(t)}}. \quad (1)$$

The *decision* the person acts on is the more probable reading, $\hat{h} = h_2$ if $b > \frac{1}{2}$ else h_1 ; “jumping to conclusions” is then simply committing to $\hat{h}$ while confidence is high but accumulated evidence is small.

L is one number that stores the whole belief, and it is deliberately *not* a yes/no. $L = 0$ means genuinely undecided; the further from 0, the more sure. Writing the belief this way lets each new piece of evidence just *add* to L , and it makes “jumping to conclusions” precise: it is L shooting far from 0 after very little has actually been added.

2.2 Metacognition is control, so belief obeys a rate law

FROM HAN & GU (2025), p. 3 ... *its core is control—i.e., stop, check, and strategy shift.*

If the essence of metacognition is *control* exercised in real time, then the belief is not computed once but continuously steered. This is exactly what a differential equation expresses: it says how the belief *changes each moment* as a function of the current belief and the incoming situation $S(t)$,

$$\frac{dL}{dt} = \mathcal{M}(L(t), S(t)), \quad \text{so that} \quad L(t) = L(0) + \int_0^t \mathcal{M}(L(\tau), S(\tau)) d\tau. \quad (2)$$

The chapter also tells us what the controller *listens to*: Proust’s procedural signals.

FROM HAN & GU (2025), p. 2 ... *monitors and corrects ongoing cognitive processing via metacognitive feelings such as confidence, fluency, and effort.*

I take confidence to be the readable procedural signal, and define it as distance from the undecided point,

$$c(t) = |2b(t) - 1| \in [0, 1]. \quad (3)$$

Equation (2) is just: “the belief now = the belief a moment ago + the small push metacognition applied.” Adding up all those small pushes over time is the integral. Nothing here assumes the pushes are large or fast; that is decided by the ingredients we plug into $\mathcal{M}$ next. Confidence (3) is 0 when unsure and 1 when certain either way.

2.3 The self–context dial (the cognitive referencing system)

FROM HAN & GU (2025), p. 7 ... *a cognitive referencing system that tilts toward either an internal (self-reflective) or an external (context/other-oriented) anchor is prone to systematic distortions... a dynamic regulator moving between two axes—self-awareness and context/other-awareness—to enable early detection and adjustment of imbalances.*

This sentence gives the model its central moving part. Evidence arrives on two channels, and the controller *weights* them. Let $\alpha(t) \in [0, 1]$ be the weight on the internal channel (so $1 - \alpha$ is the weight on the external channel; on the simplex these are the two coordinates $\alpha_{\text{self}}, \alpha_{\text{context}}$ that you wrote, but one number suffices). The two channels are:

$$\text{internal (affect / introspection): } \lambda_S(L) = a_S + r \sigma(L), \quad (4)$$

$$\text{external (situational cues): } \lambda_C(t) = a_C + \xi(t), \quad (5)$$

where $a_S \geq 0$ is a baseline threat-leaning affect, $a_C \leq 0$ says cues tend to *disconfirm* the threat reading, and $\xi(t)$ is time-correlated noise standing for the ambiguity of real situations. The term $r \sigma(L)$ is the crucial one and is justified in §2.5.

α is a volume knob between two inputs: your own gut/affect on one side, the room’s cues on the other. Turn it up and inner feelings dominate the belief; turn it down and outside cues do. The chapter’s “tilt” *is* this knob sitting off-center.

2.4 Concurrency: a coupled system, not a staircase

FROM HAN & GU (2025), ABSTRACT / P. 2 ... *a dynamic regulatory capacity that coordinates both axes concurrently and interactively, preventing drifts toward over-internalization ... and over-externalization.*

Because the two axes are *co-regulated* rather than sequenced, belief and dial evolve *together*. Combining the weighted channels with the leak of §1, and letting the dial relax toward a personal set-point while being nudged by the therapist, gives the coupled pair that is the whole model:

$$\frac{dL}{dt} = \underbrace{\alpha \lambda_S(L)}_{\text{self}} + \underbrace{(1 - \alpha) \lambda_C(t)}_{\text{context}} - \kappa L, \quad (6)$$

$$\frac{d\alpha}{dt} = -\gamma (\alpha - \alpha_0) + u(t). \quad (7)$$

The leak $-\kappa L$ implements the chapter’s “controlled re-appraisal” and its notion of *calibration* (footnote 1, p. 3: adjusting judgment so confidence tracks evidence): absent fresh evidence, the belief drifts back toward equipoise. (Note: in your draft both lines of the coupled system were typeset as dX/dt ; the second is the dial’s equation, $d\alpha/dt$, as above.)

Read (6) left to right: the belief is pushed by inner signals *times the knob*, plus outer cues *times one-minus-the-knob*, minus a small pull back toward “unsure.” Read (7): the knob drifts back to the person’s habitual setting α_0 , except when a therapist nudge u moves it. Both happen at the same time—that is the “concurrent” in your abstract.

2.5 Over-internalization is a positive-feedback loop

FROM HAN & GU (2025), P. 4–5 ... *over-internalization (excessive self-focus) appears as amplified rumination and guilt or shame, with a preoccupation with “finding the cause within,” thereby blocking corrective input from situational cues. . . an instance of calibration failure in metacognitive feelings where fluency is mistaken for validity.*

“Finding the cause within” and “fluency mistaken for validity” describe a loop: the more one already leans toward the self-referential reading, the more one’s own affect is taken as further evidence *for* it. That is precisely the term $r\sigma(L)$: internal evidence grows with current belief. When the dial α is high, this loop can overwhelm the disconfirming external channel and the leak, and L runs away—a consolidated, cue-insensitive threat belief. This is the mathematical signature of a delusion becoming fixed.

The dangerous ingredient is $r\sigma(L)$: “the surer I already am, the more my feelings confirm it.” With the knob turned toward the inside, this becomes a snowball—each turn adds to itself—and outside reassurance can no longer get in. Turn the knob back down and the

snowball stops, which is exactly what the pivot does below.

2.6 Over-externalization is tracking noisy cues

FROM HAN & GU (2025), p. 5 ... *overly drawn to others' expressions or behaviors or to rumors while neglecting internal monitoring... rendering responses erratic across situations.*

When α is small, (6) is dominated by $(1 - \alpha)\lambda_C(t)$, i.e. by the *noisy* external channel. The belief then follows the moment-to-moment fluctuation of whichever cue is salient—“erratic across situations”—with weak internal anchoring. No separate mechanism is needed; erraticism falls out of a low knob feeding on ambiguous input.

2.7 Pivots and micro-cycles are the control input $u(t)$

FROM HAN & GU (2025), p. 6, FOOTNOTE 3 ... *when imbalance signals—over-internalization or over-externalization—are detected, immediately switches the rule-set between self- and context-work to recenter the trajectory within the adaptive corridor.*

This defines $u(t)$ as *feedback* control: it reads an imbalance signal and pushes the dial the other way. A minimal law, faithful to Pivot A (over-internalization $\rightarrow$ context-work) and Pivot B (over-externalization $\rightarrow$ self-work), is

$$u(t) = \begin{cases} -p & \text{if } b > \theta_H \text{ and } \dot{L} > 0 & (\text{Pivot A: consolidating} \Rightarrow \text{knob out}) \\ +p & \text{if } \alpha < \theta_L \text{ and } |\dot{b}| \text{ large} & (\text{Pivot B: cue-driven} \Rightarrow \text{knob in}) \\ 0 & \text{otherwise} & (\text{inside the corridor: do nothing}). \end{cases} \quad (8)$$

The 5-step micro-cycle (footnote 2, p. 6) is the repeated application of this loop: rate confidence $\rightarrow$ generate alternatives $\rightarrow$ weigh evidence $\rightarrow$ reweight cues $\rightarrow$ recalibrate confidence, on the order of 30–180 s (p. 9). In the implementation the thresholds are split into engage/release pairs ($\theta_H^{\text{on}} > \theta_H^{\text{off}}$), a hysteresis (Schmitt-trigger) latch that prevents the pivot from chattering at the boundary; (8) shows the logic in its simplest form.

u is the therapist watching the dial. If the belief is snowballing inward, u turns the knob *outward* (list the room's cues, take a third-person view). If the person is blown about by every cue, u turns it *inward* (separate feeling from thought). When things are balanced, u does nothing—the corridor is where no correction is needed.

2.8 The four control parameters are already in the chapter

Strikingly, the chapter lists the very parameters this model needs:

FROM HAN & GU (2025), p. 9 ... (1) *confidence gain—the rate at which confidence rises relative to evidence*, (2) *ambiguity threshold*... (3) *cue weighting—relative allocation to internal versus external cues*, and (4) *switch sensitivity—the ease stop-check-shift routines can be invoked*.

Each maps to a knob in (6)–(8) (Table 2), which is strong evidence that the formalization is tracking the chapter rather than importing outside structure. The between-session “baseline tilt (set-point) and regulatory gain” (p. 8) are α_0 and γ ; the alignment with Jung's introversion–extraversion (p. 8) is the observation that α_0 is a personality parameter.

Table 2: The chapter’s control parameters (p. 9) map one-to-one onto the model.

Chapter’s parameter (p. 9)	Model quantity	Plain meaning
confidence gain	small κ / fast $ L $ growth	how fast certainty outruns evidence
ambiguity threshold	tolerance of ξ before commit	how long you can sit with not-knowing
cue weighting	α	inner-vs-outer allocation (the dial)
switch sensitivity	pivot gain p , thresholds θ_H, θ_L	how easily a pivot fires
baseline tilt (p. 8)	α_0	the habitual lean (a personality trait)
regulatory gain (p. 8)	γ	how strongly the dial returns home

3 Simulation experiments

I now ask whether the system (1)–(8) *behaves* as the chapter says people behave. All runs use the single “they laughed at me” scenario from the clinical vignette (p. 12); parameters are in Appendix A, code in Appendix B.

3.1 Experiment 1 — three regimes from one personality set-point

Claim tested (p. 7): a system that “tilts toward either an internal... or an external... anchor is prone to systematic distortions.” **Setup:** vary only the set-point $\alpha_0 \in \{0.85, 0.5, 0.15\}$; everything else fixed. **Result** (Fig. 1): with $\alpha_0 = 0.85$ the ruminative loop $r\sigma(L)$ drives the threat belief to ≈ 0.98 with high confidence, ignoring the disconfirming cue (delusion-like fixity); with $\alpha_0 = 0.5$ the belief stays near equipoise at low confidence (ambiguity tolerated); with $\alpha_0 = 0.15$ the belief is low but visibly erratic, tracking cue noise—matching “responses erratic across situations” (p. 5). One trait parameter reproduces all three described clinical pictures.

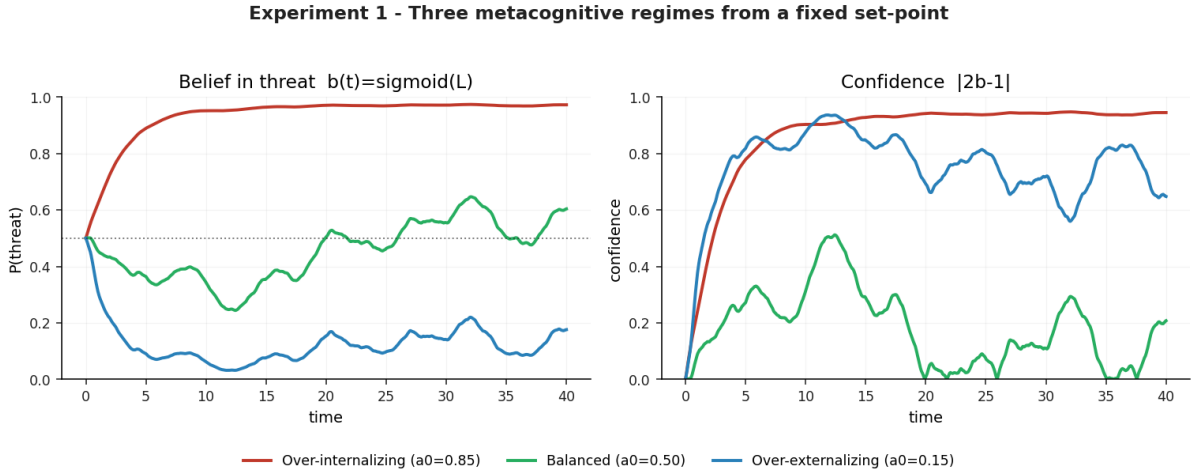


Figure 1: Belief $b(t)$ and confidence $c(t)$ under the three set-points. Over-internalization consolidates; balance stays uncertain; over-externalization is erratic.

3.2 Experiment 2 — jumping to conclusions as premature confidence

Claim tested (p. 5): “feeling = fact... leading to premature closure (JTC).” **Setup:** a high-gain configuration (small leak κ , high α_0) vs. a cautious one; commitment when c crosses a threshold $c^* = 0.8$. **Result** (Fig. 2): the high-gain agent crosses c^* after very little evidence has accumulated, while the cautious agent never over-commits. JTC appears here not as an added rule but as confidence outrunning evidence—i.e. a large “confidence gain” (p. 9).

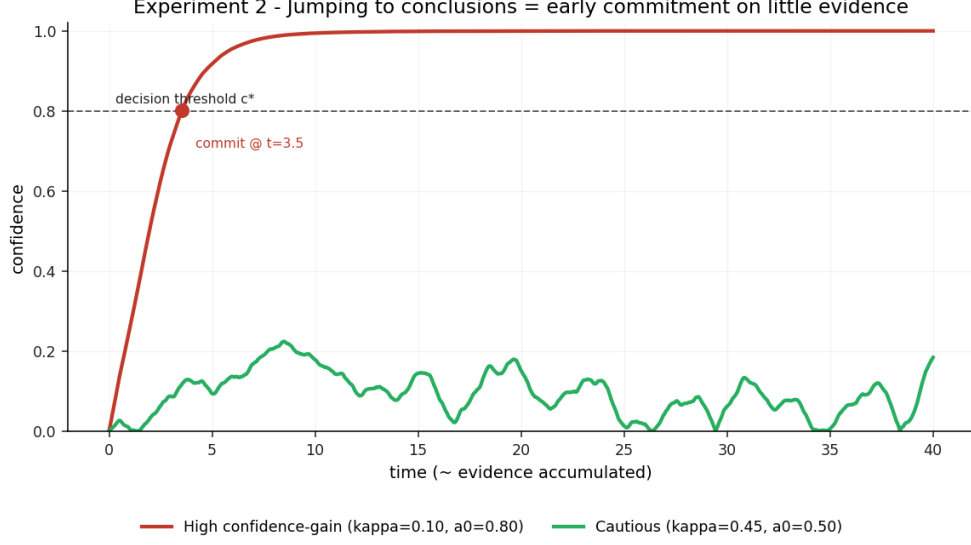


Figure 2: JTC as early threshold-crossing: high confidence-gain commits on little evidence; the cautious agent stays below the decision line.

3.3 Experiment 3 — a pivot rescues a consolidating belief

Claim tested (p. 6): pivots “recenter the trajectory within the adaptive corridor.” **Setup:** the over-internalizing patient of Exp. 1, run with the (hysteretic) pivot law (8) on vs. off, same noise seed. **Result** (Fig. 3): without intervention the belief locks at ≈ 0.97 ; with the pivot, once b rises past the engage threshold θ_H^{on} the dial α is pushed down, disconfirming cues re-enter, and the belief is pulled back toward the corridor, releasing only when b falls below θ_H^{off} . The intermittent firing of α is the repeated micro-cycle—the model’s picture of *why* the loop must be brief and iterated (p. 5)—while the hysteresis band keeps it from chattering.

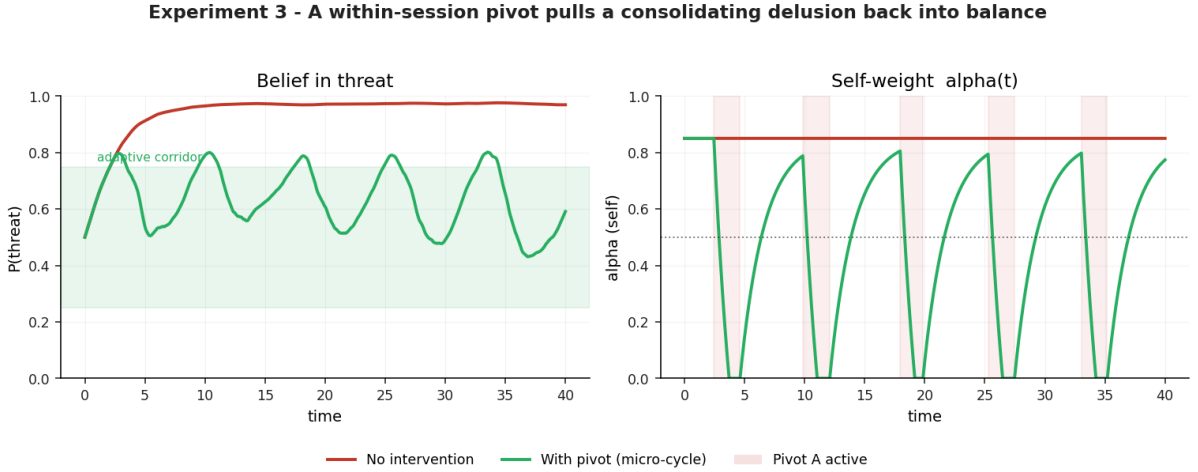


Figure 3: With vs. without the pivot. Left: the intervened belief stays out of consolidation. Right: $\alpha(t)$ shows Pivot A firing repeatedly (shaded).

3.4 Experiment 4 — the adaptive corridor as an attractor

Claim tested (p. 7): “A diagonal ‘adaptive corridor’ indicates balanced co-regulation, with a pivot to self-work at the upper left and to context-work at the lower right.” **Setup:** the

self/context engagement plane of your Figure 2, with a vector field that relaxes toward the diagonal and pivots that push out of the two risk corners. **Result** (Fig. 4): the corridor becomes a genuine *attractor*—trajectories from both risk corners are drawn back into the band—turning the chapter’s static map into dynamics.

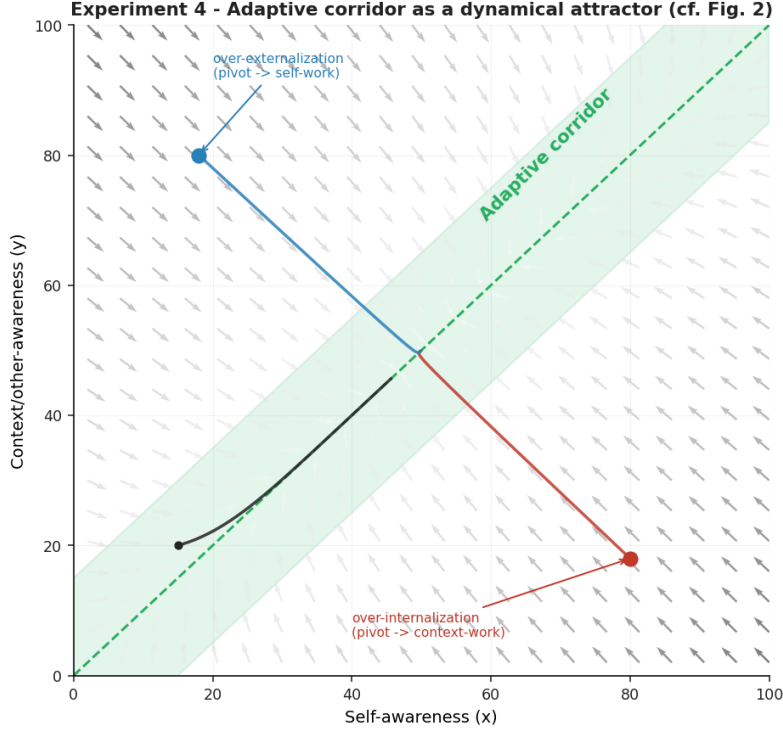


Figure 4: The adaptive corridor rendered as a dynamical attractor in the self/context plane (cf. Fig. 2 of the chapter).

3.5 Experiment 5 — the two minimal indices are diagnostic of balance

Claim tested (p. 6): tracking *confidence–outcome congruence* and *tolerance of ambiguity* suffices to read out balance. **Setup:** for each regime I run a truth-balanced ensemble (the hidden state is threat or benign with equal probability) and, at commitment, record confidence and correctness; congruence is summarized by the Expected Calibration Error (ECE, the average gap between confidence and accuracy). Separately, I present a genuinely ambiguous stimulus (internal and external evidence cancel, elevated noise) and measure how long the agent stays with it before confidence crosses a commit level. **Result** (Fig. 5): the balanced regime is best calibrated (lowest ECE, its reliability curve hugging the diagonal) and the most ambiguity-tolerant; the over-internalizer is confidently wrong (large ECE—the chapter’s “persistent overconfidence → suspect internal excess,” p. 12), and the over-externalizer commits fastest on noise (low tolerance—“jumping-to-conclusions,” p. 12). The chapter’s two low-burden indices thus separate the regimes on their own.

3.6 Experiment 6 — concurrent co-regulation beats a gated sequence

Claim tested (Table 1, p. 4; §2.4): concurrent co-regulation should outperform a strict “self → context” sequence. **Setup:** on the same substrate and noise, I compare a *sequential* policy (the set-point $\alpha_0(t)$ is gated: high, i.e. self-work, for the first half; low, i.e. context-work, for the second half; no feedback) against the *balanced* policy (the hysteretic pivot). A burst of threat-consistent evidence is injected mid-run to probe recovery. I score corridor dwell-time

Experiment 5 - The balanced regime scores best on both minimal indices

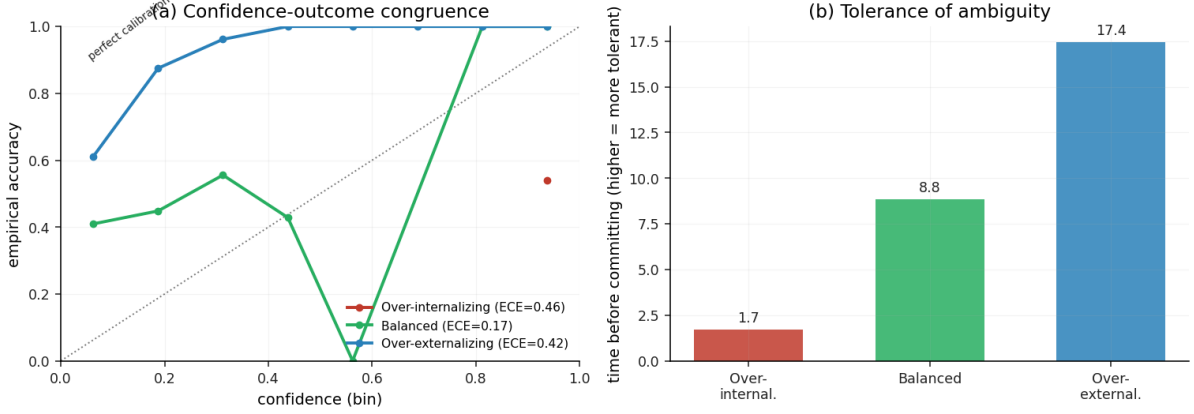


Figure 5: The paper’s minimal indices. (a) Reliability diagram with ECE per regime; balance is best calibrated. (b) Tolerance of ambiguity; balance stays with not-knowing longest.

(fraction of time the belief stays in the balanced band) and recovery time after the perturbation. **Result** (Fig. 6): the balanced policy spends far more time in the corridor (dwell 0.84 vs. 0.12) and re-enters it faster after the shock (recovery 3.5 vs. 5.9). A fixed schedule is repeatedly “in the wrong mode” for the moment’s demand; state-dependent co-regulation is not—exactly the chapter’s argument, now quantified.

Experiment 6 - Concurrent co-regulation beats a gated sequence

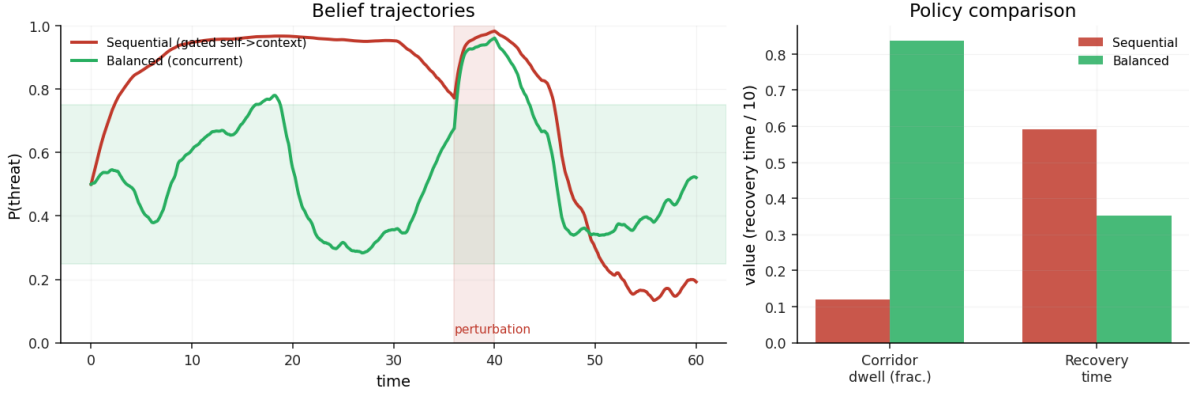


Figure 6: Sequential (gated) vs. balanced (concurrent) policy under a mid-run perturbation. Left: belief trajectories with the corridor and perturbation shaded. Right: corridor dwell-fraction and (scaled) recovery time.

4 Discussion

What the formalization buys. Three of the chapter’s claims are, in the model, not assumptions but *consequences*. Over-internalization consolidating into fixity, over-externalization being erratic, and JTC being premature confidence all emerge from the single coupled system once the ruminative feedback term $r\sigma(L)$ and the dial α are in place. The strongest internal check is Table 2: the four control parameters the model requires are the four the chapter already names on p. 9. Experiments 5–6 add quantitative support: the chapter’s two minimal indices separate

the regimes, and its verbal preference for concurrency over sequence becomes a measurable gap in corridor dwell-time and recovery speed.

Limitations and honest caveats. This is one scenario with two hypotheses; nothing here speaks to hallucinations or to multi-agent social settings. The pivot law (8) is a caricature of a clinician’s judgment. And, as the chapter itself stresses (p. 10, §4.1), an immediate in-session effect is *not* lasting metacognitive change; the model’s fast dynamics should not be over-read as the slow internalization the chapter cares about.

What was refined here, and what remains. Two refinements flagged in an earlier draft are now in the model. The pivot uses a *hysteresis* band (separate engage/release thresholds $\theta_H^{\text{on}}, \theta_H^{\text{off}}$), which removes the chattering seen with a bare threshold and reads as a clinician *committing* to a rebalancing move rather than flickering at the boundary. And interpretation is kept as a posterior while an explicit *commitment* read-out (the $\arg \max$ at decision time) is used to compute the Exp. 5 indices. Remaining work is empirical rather than structural: fitting the parameters of Table 2 to real confidence–outcome and ambiguity-tolerance logs, and extending beyond a single two-hypothesis scene.

Status. Following the chapter, this is offered as a conceptual/quantitative *interpretation*, a candidate formal specification of the kind the authors reserved for future work—not a validated model. Its value is as a shared, runnable object we can now poke at together.

A Collected model and parameters

The full system, for reference:

$$\begin{aligned} \dot{L} &= \alpha(a_S + r\sigma(L)) + (1 - \alpha)(a_C + \xi(t)) - \kappa L, & \dot{\alpha} &= -\gamma(\alpha - \alpha_0) + u(t), \\ d\xi &= -\theta\xi dt + \sigma_C dW, & c &= |2\sigma(L) - 1|, \end{aligned}$$

with $u(t)$ from (8). Baseline parameters: $a_S = 0.20$, $r = 1.20$, $a_C = -0.70$, $\sigma_C = 0.50$, $\theta = 1.0$, $\kappa = 0.30$, $\gamma = 0.50$, pivot gain $p = 0.9$, hysteresis $\theta_H^{\text{on}} = 0.78$, $\theta_H^{\text{off}} = 0.62$, $\theta_L = 0.25$. Set-point α_0 is varied per experiment; Exp. 2 uses $\kappa \in \{0.10, 0.45\}$. Exp. 5 runs a truth-balanced ensemble (evidence set to favor threat or benign per trial) and an ambiguous stimulus ($a_S = 0.40$, $a_C = -0.40$, $\sigma_C = 0.70$). Exp. 6 drives $\alpha_0(t)$ as a gated schedule for the sequential policy and injects a threat-evidence burst (magnitude 1.6 over a short window) to probe recovery.

B Reference implementation

Figures are produced by the `mcbalance` package (`model.py`, `metrics.py`, `experiments.py`); see the project repository for the full source, metrics, and a one-command reproduction script. The reusable core dynamics (explicit Euler–Maruyama, $dt = 0.02$, hysteretic pivot latch):

```

1  """Core dynamical model of the Metacognitive Balance perspective.
2
3  Formalizes Han & Gu (2025), "Toward Metacognitive Balance: A Framework for
4  Cognitive and Social Functioning in Schizophrenia" (IntechOpen,
5  doi:10.5772/intechopen.1012821) as a coupled control system.
6
7  State
8  ----
9  L(t) : log-odds of the threat / self-referential hypothesis h2 vs benign h1.
10         belief b = sigmoid(L) in [0, 1].
11  alpha(t) : self-weight in [0, 1]. alpha_self = alpha, alpha_context = 1 - alpha.
12
13  Dynamics

```

```

14 -----
15     dL/dt = alpha * lambda_S(L) + (1 - alpha) * lambda_C(t) - kappa * L + P_ext(t)
16     dalpha/dt = -gamma * (alpha - alpha0(t)) + u(t)
17
18 with
19     lambda_S(L) = aS + r * sigmoid(L) # internal affect + ruminative self-amplification
20     lambda_C(t) = aC + xi(t) # external cues (disconfirming) + OU ambiguity noise
21     u(t) # pivot control law, with hysteresis (see PivotLatch)
22     P_ext(t) # optional external evidence perturbation
23
24 Everything is deliberately minimal so the qualitative behaviour can be read off
25 the equations. See "experiments.py" for the studies and "metrics.py" for the
26 paper's two "minimal indices".
27 """
28 from __future__ import annotations
29
30 from dataclasses import dataclass, field
31 from typing import Callable, Optional, Tuple, Dict
32
33 import numpy as np
34
35 __all__ = ["Params", "sigmoid", "confidence", "simulate", "PivotLatch"]
36
37 DEFAULT_SEED = 20250701
38
39
40 def sigmoid(x: np.ndarray | float) -> np.ndarray | float:
41     """Logistic squashing function mapping log-odds to a probability in [0, 1]."""
42     return 1.0 / (1.0 + np.exp(-x))
43
44
45 def confidence(L: np.ndarray | float) -> np.ndarray | float:
46     """Metacognitive confidence: distance of the belief from equipoise, in [0, 1]."""
47     return np.abs(2.0 * sigmoid(L) - 1.0)
48
49
50 @dataclass
51 class Params:
52     """Model parameters. Field names mirror the symbols used in the manuscript.
53
54     The four control parameters that Han & Gu (2025, p. 9) enumerate map onto
55     these fields: confidence gain -> kappa (leak); ambiguity threshold -> how
56     long |L| stays small under sigC noise; cue weighting -> alpha; switch
57     sensitivity -> p_gain / thresholds.
58     """
59     # --- evidence channels ---
60     aS: float = 0.20 # baseline threat-leaning internal affect
61     r: float = 1.20 # ruminative self-amplification gain ("finding the cause within")
62     aC: float = -0.70 # external cues disconfirm the threat reading on average
63     sigC: float = 0.50 # ambiguity: amplitude of external (OU) noise
64     theta_ou: float = 1.0 # OU mean-reversion rate (noise time-correlation)
65
66     # --- integration / regulation ---
67     kappa: float = 0.30 # leak toward equipoise (controlled re-appraisal / calibration)
68     gamma: float = 0.50 # CRS homeostatic return rate to the set-point
69     alpha0: float = 0.50 # personality set-point (baseline tilt); Jung intro/extraversion
70
71     # --- pivot control (with hysteresis) ---
72     pivot: bool = False
73     p_gain: float = 0.90 # pivot strength (switch sensitivity)
74     theta_H_on: float = 0.78 # Pivot A engages when belief b exceeds this
75     theta_H_off: float = 0.62 # ... and disengages only when b falls below this (hysteresis)
76     theta_L: float = 0.25 # Pivot B considered when alpha below this
77     erratic_on: float = 0.9 # Pivot B engages when |db/dt| exceeds this
78     erratic_off: float = 0.4 # ... and disengages when |db/dt| falls below this
79
80
81 class PivotLatch:
82     """Stateful, hysteretic pivot trigger.
83
84     A bare threshold makes the pivot *chatter* (rapidly toggle) near the
85     boundary. Separate on/off thresholds (Schmitt-trigger style) latch the pivot
86     so it engages decisively and releases only once the imbalance has clearly

```

```

87     resolved -- a more clinically realistic "commit to the rebalancing move".
88     """
89
90     def __init__(self, P: Params):
91         self.P = P
92         self.A_on = False # Pivot A (over-internalization -> context-work)
93         self.B_on = False # Pivot B (over-externalization -> self-work)
94
95     def __call__(self, b: float, dL: float, alpha: float, db: float) -> float:
96         P = self.P
97         # --- Pivot A: over-internalization -> push knob toward context (alpha down) ---
98         if self.A_on:
99             if b < P.theta_H_off:
100                 self.A_on = False
101             elif b > P.theta_H_on and dL > 0:
102                 self.A_on = True
103
104         # --- Pivot B: over-externalization -> push knob toward self (alpha up) ---
105         if self.B_on:
106             if abs(db) < P.erratic_off:
107                 self.B_on = False
108             elif alpha < P.theta_L and abs(db) > P.erratic_on:
109                 self.B_on = True
110
111         if self.A_on:
112             return -P.p_gain
113         if self.B_on:
114             return +P.p_gain
115         return 0.0
116
117
118     def simulate(
119         P: Params,
120         T: float = 40.0,
121         dt: float = 0.02,
122         L0: float = 0.0,
123         alpha_init: Optional[float] = None,
124         seed: int = DEFAULT_SEED,
125         alpha0_fn: Optional[Callable[[float], float]] = None,
126         perturb: Optional[Tuple[float, float, float]] = None,
127     ) -> Dict[str, np.ndarray]:
128         """Integrate the coupled system with the explicit Euler-Maruyama scheme.
129
130         Parameters
131         -----
132         P : Params
133             Model parameters.
134         T, dt : float
135             Total time and step. Time is in the same abstract units as the
136             30-180 s micro-cycle discussed by Han & Gu (2025, p. 9).
137         L0 : float
138             Initial log-odds (0.0 = equipoise).
139         alpha_init : float, optional
140             Initial self-weight. Defaults to 'P.alpha0' (or 'alpha0_fn(0)').
141         seed : int
142             RNG seed for the external ambiguity noise.
143         alpha0_fn : callable, optional
144             Time-varying set-point 'alpha0(t)'. Used to express a *scheduled*
145             (sequential) policy, e.g. self-work first then context-work. When
146             supplied it overrides the constant 'P.alpha0' in the CRS equation.
147         perturb : (t0, t1, mag), optional
148             Adds a constant external-evidence rate 'mag' to dL for 't in [t0, t1]'
149             -- a burst of (dis)confirming evidence used to probe recovery.
150
151         Returns
152         -----
153         dict of np.ndarray with keys: t, L, alpha, b, conf, xi, u.
154         """
155         rng = np.random.default_rng(seed)
156         n = int(round(T / dt))
157         t = np.linspace(0.0, T, n)
158
159         L = np.zeros(n)

```

```

160 A = np.zeros(n)
161 xi = np.zeros(n)
162 U = np.zeros(n)
163
164 set_point = (lambda _t: P.alpha0) if alpha0_fn is None else alpha0_fn
165 L[0] = L0
166 A[0] = set_point(0.0) if alpha_init is None else alpha_init
167
168 latch = PivotLatch(P)
169 prev_b = sigmoid(L[0])
170
171 for k in range(n - 1):
172     # Ornstein-Uhlenbeck ambiguity noise on the external channel
173     xi[k + 1] = xi[k] - P.theta_ou * xi[k] * dt + P.sigC * np.sqrt(dt) * rng.standard_normal()
174
175     lamS = P.aS + P.r * sigmoid(L[k])
176     lamC = P.aC + xi[k]
177     p_ext = 0.0
178     if perturb is not None and perturb[0] <= t[k] < perturb[1]:
179         p_ext = perturb[2]
180
181     dL = A[k] * lamS + (1.0 - A[k]) * lamC - P.kappa * L[k] + p_ext
182
183     b = sigmoid(L[k])
184     db = (b - prev_b) / dt
185     prev_b = b
186
187     u = latch(b, dL, A[k], db) if P.pivot else 0.0
188     U[k] = u
189
190     dA = -P.gamma * (A[k] - set_point(t[k])) + u
191
192     L[k + 1] = L[k] + dL * dt
193     A[k + 1] = float(np.clip(A[k] + dA * dt, 0.0, 1.0))
194
195     return dict(t=t, L=L, alpha=A, b=sigmoid(L), conf=confidence(L), xi=xi, u=U)

```

References

Mihwa Han and Hoedeok Gu. Toward Metacognitive Balance: A Framework for Cognitive and Social Functioning in Schizophrenia. In *Metacognition in Action - Insights into Learning, AI, and Self-Regulation*. IntechOpen, October 2025. ISBN 978-0-85466-727-7 978-0-85466-728-4 978-0-85466-729-1. doi: 10.5772/intechopen.1012821.